

Phase 2 Setup Plan: pberp.duckdns.org for Haritha Hospitals

Created: 2026-08-20 14:46 IST (revised 15:46 IST per user decision) **Owner:** Venkat (Processbricks) **Target:** New Frappe site at pberp.duckdns.org **Compose project name:** pberp **Stack:** Frappe 16.x + ERPNext 16.x + HRMS **16.5.0 pinned** (Lesson #44) + payments **Estimated time:** 5-7 hours total (simplified from custom image build approach)

Approach: Vanilla image + runtime app install (per user preference)

User chose NOT to build a custom Docker image. Use the vanilla frappe/erpnext:v16 image. Install apps at runtime via bench get-app + bench install-app.

Workflow: 1. Use vanilla image: frappe/erpnext:v16 (or specific version) 2. Create site (no app install) 3. bench get-app + bench install-app for each app

Trade-off: Simpler setup (no docker build), but apps are installed at runtime rather than baked into the image. For production, custom image builds are preferred (frappe_docker canonical). For this project, vanilla image is acceptable per user.

HRMS pin: Use --branch version-15 per Lesson #44 (v16.5.0). Even on vanilla image, this pin is enforced.

Summary

Phase	What	Time	Status
A	Pre-flight (DNS, SSL, dirs)	30m	pending
B	Compose up (vanilla image)	30m	pending
C	Site creation (no app install)	15m	pending
D	Get apps + install + restart workers	30m	pending
E	Custom fields fixtures (7 fields)	30m	pending
F	MySQL grants for scheduler (Lesson #87)	5m	pending
G	Company + Holiday List	15m	pending
H	Data import (19 CSVs)	60m	pending
I	Backup setup (hardened prod_backup.sh pattern)	30m	pending
J	nginx routing (real path fix)	30m	pending
K	Test + Sign-off (100% green + delete + report)	3-4h	pending
L	Cert expiry monitoring	15m	pending

7 User Decisions (FINAL)

1. Compose project name: **pberp**
2. HRMS pin: **16.5.0** (Lesson #44 — --branch version-15 enforced even on vanilla image)
3. Ports: NO host ports published (match existing erp-dev/qa/prod convention; Docker DNS only)
4. Leave Allocation (1,470 rows): DEFER — HRMS auto-allocates on first Leave Application

5. nginx config: REAL path is /home/vijay/nginx-erp.conf (single-file bind-mount to container /etc/nginx/conf.d/erp.conf)
6. Backup: match existing (cron 0 */6 * * *, KEEP_DAYS=7 local, offsite to venkat@135.125.196.35 retains forever)
7. Test gating: 100% green vendor tests REQUIRED for sign-off → delete all test data → generate sign-off report

7 Custom Fields (Rule #9: must be in fixtures, NEVER dialog-added)

Entity	Field	Type
Shift Type	is_onscall	Check
Shift Type	is_emergency	Check
Employee	is_synthetic_data	Check
Attendance	late_entry_by	Int
Attendance	early_out_by	Int
Attendance	is_wfh	Check
Employee Checkin	is_off	Check

Probe + Research Findings (embedded)

- **frappe-only** compose project: REMOVED at 11:42 IST Aug 20 (3 containers gone)
- **Port convention:** No existing env publishes host ports. Match this.
- **nginx config:** Real served file = /home/vijay/nginx-erp.conf
- **SSL certs:** Located at /home/vijay/erpnext/ssl-certs/<subdomain>/{fullchain,privkey}.pem. **Existing certs expire 2026-10-25.** No auto-renewal script found.
- **Stale config flag:** Dead pberpfrappe.duckdns.org block in nginx config — REMOVE during Phase J
- **Backup scripts:** /home/vijay/scripts/prod_backup.sh (158 lines, hardened 2026-08-18 with set -euo pipefail + timeout 900 + PIPESTATUS[0]) — use as template
- **Cron schedule:** 0 */6 * * * for envs; erpclaw at 0 3 * * *
- **DuckDNS config:** Token at /home/vijay/erpnext/ssl-certs/duckdns-credentials.ini

Prerequisites

- DNS: pberp.duckdns.org resolves to nginx-proxy IP
- SSL cert at /home/vijay/erpnext/ssl-certs/pberp/
- Workspace files ready at /root/.openclaw/workspace/projects/haritha-hospitals/
- frappe-only removed (11:42 IST Aug 20)
- Pre-existing compose projects intact: erp-dev, erp-qa, erp-prod, erp_custom_apps, frappeclaw, nginx-proxy

Phase A: Pre-flight (~30 min)

Goal: DNS verified, SSL cert provisioned, site folder created

1. DNS verify

dig +short pberp.duckdns.org

2. SSL cert directory

ssh vijay@144.217.163.228 "sudo mkdir -p /home/vijay/erpnext/ssl-certs/pberp"

```
# 3. Provision cert (follow duckdns-credentials.ini pattern from pberpqa)
ssh vijay@144.217.163.228 "ls /home/vijay/erpnext/ssl-certs/"
```

```
# 4. Site folder
```

```
ssh vijay@144.217.163.228 "sudo mkdir -p /home/vijay/erp-pberp"
```

Verify: dig +short pberp.duckdns.org returns nginx-proxy IP; SSL cert files exist; site folder ready.

Phase B: Compose up (vanilla image) (~30 min)

Goal: 9 containers Up using vanilla frappe/erpnext:v16 image (NO custom build)

```
# 1. Copy template from erp-qa
```

```
ssh vijay@144.217.163.228 "cd /home/vijay/erp-qa && sudo tar cf - --exclude=.git --exclude="
```

```
# 2. Rename compose project + containers
```

```
ssh vijay@144.217.163.228 "cd /home/vijay/erp-pberp && sudo sed -i 's/name: erp-qa/name: p
```

```
ssh vijay@144.217.163.228 "cd /home/vijay/erp-pberp && sudo sed -i 's/erp-qa-/erp-pberp-/g
```

```
# 3. Disable custom image (comment out CUSTOM_IMAGE + CUSTOM_TAG to use vanilla)
```

```
ssh vijay@144.217.163.228 "cd /home/vijay/erp-pberp && \
  sudo sed -i 's|^CUSTOM_IMAGE=.*|# CUSTOM_IMAGE=|' .env && \
  sudo sed -i 's|^CUSTOM_TAG=.*|# CUSTOM_TAG=|' .env"
```

```
# 4. Verify docker-compose.yml uses vanilla image (look for `image:` field)
```

```
ssh vijay@144.217.163.228 "grep '^[:space:]*image:' /home/vijay/erp-pberp/docker-compose
```

```
# Should show: image: frappe/erpnext:v16 (or similar)
```

```
# 5. Update SITE_NAME in .env
```

```
ssh vijay@144.217.163.228 "sudo sed -i 's/SITE_NAME=.* /SITE_NAME=pberp.duckdns.org/' /home
```

```
# 6. Start containers (NO host ports per convention)
```

```
ssh vijay@144.217.163.228 "cd /home/vijay/erp-pberp && sudo docker compose -p pberp --env-
```

```
# 7. Wait + verify
```

```
sleep 60
```

```
ssh vijay@144.217.163.228 "docker ps | grep pberp" # Should show 9 containers
```

```
ssh vijay@144.217.163.228 "docker inspect --format '{{.State.Health.Status}}' erp-pberp-ba
```

Verify: 9 containers Up, backend healthy, image is frappe/erpnext:v16 (vanilla).

Phase C: Site creation (~15 min)

```
ssh vijay@144.217.163.228 "cd /home/vijay/erp-pberp && \
  docker exec -u frappe erp-pberp-backend-1 bash -c '
    cd /home/frappe/frappe-bench
    bench new-site pberp.duckdns.org \
      --mariadb-root-password <ROOT_PWD> \
      --admin-password <ADMIN_PWD>
  '"
```

DO NOT install any app here — apps installed in Phase D via `bench get-app + bench install-app`.

Verify: UI login works at <http://pberp.duckdns.org:8000>. Only frappe is installed.

Phase D: Get apps + install + restart workers (~30 min)

Goal: Install payments + erpnext + hrms (16.5.0 pinned) on the site + restart workers per Lesson #46

1. Get apps (via runtime installation per user preference)

```
ssh vijay@144.217.163.228 "cd /home/vijay/erp-pberp && \
  docker exec -u frappe erp-pberp-backend-1 bash -c '
    cd /home/frappe/frappe-bench
    bench get-app payments
    bench get-app erpnext --branch version-16
    bench get-app hrms --branch version-15 # PINNED — Lesson #44
  '"
```

2. Install apps on site

```
ssh vijay@144.217.163.228 "cd /home/vijay/erp-pberp && \
  docker exec -u frappe erp-pberp-backend-1 bash -c '
    cd /home/frappe/frappe-bench
    bench --site pberp.duckdns.org install-app erpnext hrms payments
  '"
```

3. Verify apps installed (Lesson #80 — both DB and filesystem must match)

```
ssh vijay@144.217.163.228 "cat /home/vijay/erp-pberp/sites/apps.txt"
```

Should show 4 lines: frappe, erpnext, hrms, payments

```
ssh vijay@144.217.163.228 "cd /home/vijay/erp-pberp && \
  docker exec -u frappe erp-pberp-backend-1 bash -c '
    cd /home/frappe/frappe-bench
    bench --site pberp.duckdns.org console -c '"import frappe; print(frappe.get_installed_
  '"
```

Should show: ['frappe', 'erpnext', 'hrms', 'payments']

4. Restart workers (Lesson #46)

```
ssh vijay@144.217.163.228 "cd /home/vijay/erp-pberp && sudo docker compose -p pberp restart
sleep 30"
```

Verify: All 4 apps visible. Workers restarted. UI works.

Phase E: Custom fields fixtures (~30 min)

Pre-author fixture JSON

```
ssh vijay@144.217.163.228 "cat > /home/vijay/erp-pberp/masters/pberp_custom_fields.json <<
[
  {"doctype":"Custom Field","name":"is_oncall","dt":"Shift Type","fieldname\"
  {"doctype":"Custom Field","name":"is_emergency","dt":"Shift Type","fieldnam
  {"doctype":"Custom Field","name":"is_synthetic_data","dt":"Employee","field
  {"doctype":"Custom Field","name":"late_entry_by","dt":"Attendance","fieldna
```

```

{"doctype":"Custom Field","name":"early_out_by","dt":"Attendance","fieldname":
{"doctype":"Custom Field","name":"is_wfh","dt":"Attendance","fieldname":
{"doctype":"Custom Field","name":"is_off","dt":"Employee Checkin","fieldname":
]
EOF
"

```

Apply via console

```

ssh vijay@144.217.163.228 "docker exec -u frappe erp-pberp-backend-1 bash -c '
    cd /home/frappe/frappe-bench
    bench --site pberp.duckdns.org console <<\"PYTHON\"
import frappe, json
frappe.flags.mute_emails = True
data = json.load(open(\"/home/vijay/erp-pberp/masters/pberp_custom_fields.json\"))
for item in data:
    frappe.get_doc(dict(item)).insert(ignore_permissions=True)
frappe.db.commit()
print(f\"Inserted {len(data)} custom fields\")
PYTHON
\""

```

Verify: - frappe.get_meta("Shift Type").custom_fields length = 2 - frappe.get_meta("Attendance").length = 3 - frappe.get_meta("Employee").custom_fields length = 1 - frappe.get_meta("Employee Checkin").custom_fields length = 1

Phase F: MySQL grants (Lesson #87) — ~5 min

Critical: Run BEFORE scheduler wakes up (within ~30s of Phase B)

Find scheduler IP

```
ssh vijay@144.217.163.228 "docker inspect --format '{{range .NetworkSettings.Networks}}{{.IPAddr}}
```

Get DB credentials

```
ssh vijay@144.217.163.228 "grep -E 'MYSQL|DB_' /home/vijay/erp-pberp/.env"
```

Pre-create grant

```
ssh vijay@144.217.163.228 "docker exec erp-pberp-db-1 mariadb -u root -p<ROOT_PWD> -e \"
GRANT ALL ON pberp_db.* TO 'pberp_user'@'<SCHEDULER_IP>' IDENTIFIED BY '<DB_PWD>';
FLUSH PRIVILEGES;
\""
```

Verify no 1045 errors after 5 min

```
sleep 300
```

```
ssh vijay@144.217.163.228 "docker logs erp-pberp-scheduler-1 --since 5m | grep -c 1045" #
```

Verify: Zero 1045 errors in scheduler logs.

Phase G: Company + Holiday List (~15 min)

Copy CSVs into container workspace

```
ssh vijay@144.217.163.228 "docker cp /home/vijay/erp-pberp/masters/company.csv erp-pberp-b
ssh vijay@144.217.163.228 "docker cp /home/vijay/erp-pberp/masters/holiday_list.csv erp-pb
```

```
ssh vijay@144.217.163.228 "docker cp /home/vijay/erp-pberp/masters/holiday.csv erp-pberp-b
```

```
# Import via Data Import tool (UI) or console
# (See all_schemas.csv for field names)
```

Verify: UI shows Haritha Hospitals company + 14 holidays.

Phase H: Data import (~60 min)

Order (smallest first): 1. Department, Designation, Employment Type, Shift Type, Leave Type (small) 2. Employee (210) 3. ~~Leave Allocation~~ DEFER (Decision #4) — HRMS auto-allocates 4. Shift Assignment (5,317), Attendance (6,300), Employee Checkin (12,562) — large

```
# Per entity (template):
```

```
ssh vijay@144.217.163.228 "docker cp /home/vijay/erp-pberp/masters/<entity>.csv erp-pberp-
ssh vijay@144.217.163.228 "docker exec -u frappe erp-pberp-backend-1 bash -c '
    cd /home/frappe/frappe-bench
    bench --site pberp.duckdns.org console <<\"PYTHON\"
import frappe, csv
rows = list(csv.DictReader(open(\"/tmp/<entity>.csv\")))
for r in rows:
    frappe.get_doc(dict({\"doctype\": \"<DocType>\", **r})).insert(ignore_permissions=True)
frappe.db.commit()
print(f\"Imported {len(rows)} <DocType>\")
PYTHON
\""
```

```
# Verify after each
```

```
ssh vijay@144.217.163.228 "docker exec -u frappe erp-pberp-backend-1 bench --site pberp.du
```

Expected counts: | Entity | Count | |—| | Department | 36 | | Designation | 48 | | Employment Type | 6 | | Shift Type | 25 | | Leave Type | 7 | | Employee | 210 (194 Active, 16 Inactive) | | Shift Assignment | 5,317 | | Attendance | 6,300 | | Employee Checkin | 12,562 |

Phase I: Backup setup (HARDENED pattern) (~30 min)

Use prod_backup.sh (158 lines) as template — has set -euo pipefail + timeout 900 + PIPESTATUS[0] (Lesson #79).

```
# 1. Copy template
```

```
ssh vijay@144.217.163.228 "sudo cp /home/vijay/scripts/prod_backup.sh /home/vijay/scripts/
```

```
# 2. Edit variables
```

```
ssh vijay@144.217.163.228 "sudo sed -i 's/SITE_NAME=.* /SITE_NAME=pberp.duckdns.org/' /home
ssh vijay@144.217.163.228 "sudo sed -i 's/BACKUP_CONTAINER=.* /BACKUP_CONTAINER=erp-pberp-b
ssh vijay@144.217.163.228 "sudo sed -i 's|LOCAL_BACKUP_DIR=.*|LOCAL_BACKUP_DIR=/home/vijay
ssh vijay@144.217.163.228 "sudo sed -i 's|OFFSITE_TARGET=.*|OFFSITE_TARGET=venkat@135.125.
```

```
# 3. Syntax check + chmod
```

```
ssh vijay@144.217.163.228 "bash -n /home/vijay/scripts/pberp_backup.sh && chmod +x /home/v
```

```
# 4. Add to cron (heredoc + atomic per Lesson #77)
```

```
ssh vijay@144.217.163.228 "crontab -l > /tmp/cron.bak && \
echo '0 */6 * * * /home/vijay/scripts/pberp_backup.sh >> /home/vijay/backups/logs/pberp_
crontab /tmp/cron.bak && rm /tmp/cron.bak"
```

5. First run (sanity)

```
ssh vijay@144.217.163.228 "sudo /home/vijay/scripts/pberp_backup.sh"
```

Verify: crontab -l shows pberp backup line. First run produces valid backup file.

Phase J: nginx routing (~30 min)

REAL path: /home/vijay/nginx-erp.conf (NOT conf.d/)

1. Backup current config

```
ssh vijay@144.217.163.228 "sudo cp /home/vijay/nginx-erp.conf /home/vijay/nginx-erp.conf.b
```

*# 2. Edit /home/vijay/nginx-erp.conf – add pberp.duckdns.org server block
Use pberpqa block as template; upstream: erp-pberp-frontend-1:8080*

3. CLEANUP: remove dead pberpfrappe.duckdns.org block

4. Test + reload

```
ssh vijay@144.217.163.228 "docker exec nginx-proxy nginx -t"
```

```
ssh vijay@144.217.163.228 "docker exec nginx-proxy nginx -s reload"
```

5. Verify

```
curl -I https://pberp.duckdns.org/ # 200/302
```

Verify: nginx -t syntax OK; curl -I https://pberp.duckdns.org/ returns 200/302; UI accessible.

Phase K: Test + Sign-off (~3-4 hr)

K1: Vendor tests (100% green REQUIRED)

```
ssh vijay@144.217.163.228 "cd /home/vijay/erp-pberp && \
docker exec -u frappe erp-pberp-backend-1 bash -c '
cd /home/frappe/frappe-bench
CI=1 bench --site pberp.duckdns.org run-tests --app hrms
'"
```

Must be 100% green. If failures, investigate before proceeding.

K2-K7: Standard 8-phase (per Lesson #48)

- K2: Smoke (login + view pages)
- K3: Schema (all 19 entities exist)
- K4: CRUD (test records)
- K5: Workflow (Shift Schedule → SSA → cron → Shift Assignment creation)
- K6: Integration (Leave → Attendance status)
- K7: Regression (existing critical workflows)

K8: Delete ALL test data (per Decision #7)

```
ssh vijay@144.217.163.228 "cd /home/vijay/erp-pberp && \
  docker exec -u frappe erp-pberp-backend-1 bench --site pberp.duckdns.org console <<\ "PYT
import frappe
frappe.db.sql(\ "DELETE FROM tabShift Assignment WHERE docstatus=0\" )
frappe.db.sql(\ "DELETE FROM tabEmployee WHERE employee_number LIKE 'TEST-%'\" )
frappe.db.sql(\ "DELETE FROM tabAttendance WHERE status='Test'\" )
frappe.db.commit()
print('Test data deleted')
PYTHON
"
```

K9: Generate sign-off report

```
ssh vijay@144.217.163.228 "cat > /home/vijay/erp-pberp/sites/sign-off-report.md << 'EOF'
# Haritha Hospitals – Phase 2 Sign-off Report
**Date:** $(date)
**Site:** pberp.duckdns.org
**Compose project:** pberp
**Approach:** Vanilla frappe/erpnext:v16 + runtime app install (per user preference)

## Test Results
### Vendor Tests (K1)
- HRMS: <PASS/FAIL> (X/Y tests, Z failures)

### Data Import Counts (K3)
| Entity | Expected | Actual |
|-----|-----|-----|
| Department | 36 | ? |
| Designation | 48 | ? |
| Employee | 210 | ? |
| Shift Type | 25 | ? |
| Shift Assignment | 5,317 | ? |
| Attendance | 6,300 | ? |
| Employee Checkin | 12,562 | ? |

## Custom Fields Applied (Rule #9)
- 7 custom fields verified

## Backup Verified
- First run: <timestamp>

## nginx Verified
- https://pberp.duckdns.org/ returns 200/302

## Sign-off
- [ ] All tests 100% green
- [ ] Counts verified
- [ ] Test data deleted
- [ ] User approves

**Signed off by:** _____
**Date:** _____
```


EOF
"

Phase L: Cert expiry monitoring (~15 min)

CRITICAL: Existing certs expire 2026-10-25. No auto-renewal script found.

1. Check expiry of all certs

```
ssh vijay@144.217.163.228 "for cert in /home/vijay/erpnext/ssl-certs/*/fullchain.pem; do e
```

2. Write cert-expiry-check.sh script

```
ssh vijay@144.217.163.228 "cat > /home/vijay/scripts/cert_expiry_check.sh << 'EOF'
```

```
#!/bin/bash
```

```
set -euo pipefail
```

```
EXPIRY=$(openssl x509 -enddate -noout -in /home/vijay/erpnext/ssl-certs/pberp/fullchain.p
```

```
EXPIRY_EPOCH=$(date -d "\"$EXPIRY\"" +%s)
```

```
NOW_EPOCH=$(date +%s)
```

```
DAYS_LEFT=$(( (EXPIRY_EPOCH - NOW_EPOCH) / 86400 ))
```

```
if [ \"$DAYS_LEFT\" -lt 30 ]; then
```

```
    echo \"WARNING: pberp cert expires in $DAYS_LEFT days ($EXPIRY)\"
```

```
fi
```

```
EOF
```

```
chmod +x /home/vijay/scripts/cert_expiry_check.sh"
```

3. Add daily check to cron

```
ssh vijay@144.217.163.228 "crontab -l > /tmp/cron.bak && \"
```

```
    echo '0 8 * * * /home/vijay/scripts/cert_expiry_check.sh >> /home/vijay/scripts/cert_exp
```

```
    crontab /tmp/cron.bak && rm /tmp/cron.bak"
```

4. Calendar reminder (manual) – renew 25 days before expiry

```
echo "Set calendar reminder for 2026-10-01 to renew pberp cert"
```

Verify: crontab -l shows daily cert check. First run produces no warning (cert valid).

Open Questions / Decisions

All 7 decisions captured. No remaining open questions.

Sign-off Checklist

Before Phase 2 closure:

- ☐ A: DNS + SSL + dir ready
- ☐ B: 9 containers Up + backend healthy (vanilla image)
- ☐ C: Site created + admin password set (no app install)
- ☐ D: All 4 apps installed + workers restarted
- ☐ E: 7 custom fields verified
- ☐ F: Zero 1045 errors (Lesson #87 prevented)
- ☐ G: Company + Holidays in UI
- ☐ H: All 19 entities imported + counts match
- ☐ I: Backup cron + first run success (hardened pattern)

- ☐ J: nginx serving pberp.duckdns.org
 - ☐ K1: Vendor tests 100% GREEN
 - ☐ K2-K7: Manual tests all PASS
 - ☐ K8: Test data deleted
 - ☐ K9: Sign-off report generated
 - ☐ L: Cert expiry monitoring set up
 - ☐ Manager approval
-

Total estimated time: 5-7 hours (down from 7-10 due to simplified approach) **Sign-off requires:** All checklist items + 100% green vendor tests